

Architecture & security review package

Diagrams and inventories for security/ATO review of the Claude apps gateway deployment (AWS GovCloud `us-gov-west-1`). Everything here is generated from the templates in `cloudformation/` at the commit this file ships with — when the code and this document disagree, the code wins and this file needs a PR.

The diagrams are hand-laid-out SVGs under `docs/ato/diagrams/`, generated by `docs/ato/diagrams/generate.py` (edit the script, re-run it, and commit both). SVGs print cleanly and embed directly in submission documents; for PNG exports: `python3 -c "import cairosvg; cairosvg.svg2png(url='docs/ato/diagrams/<name>.svg', write_to='<name>.png', scale=2)".`

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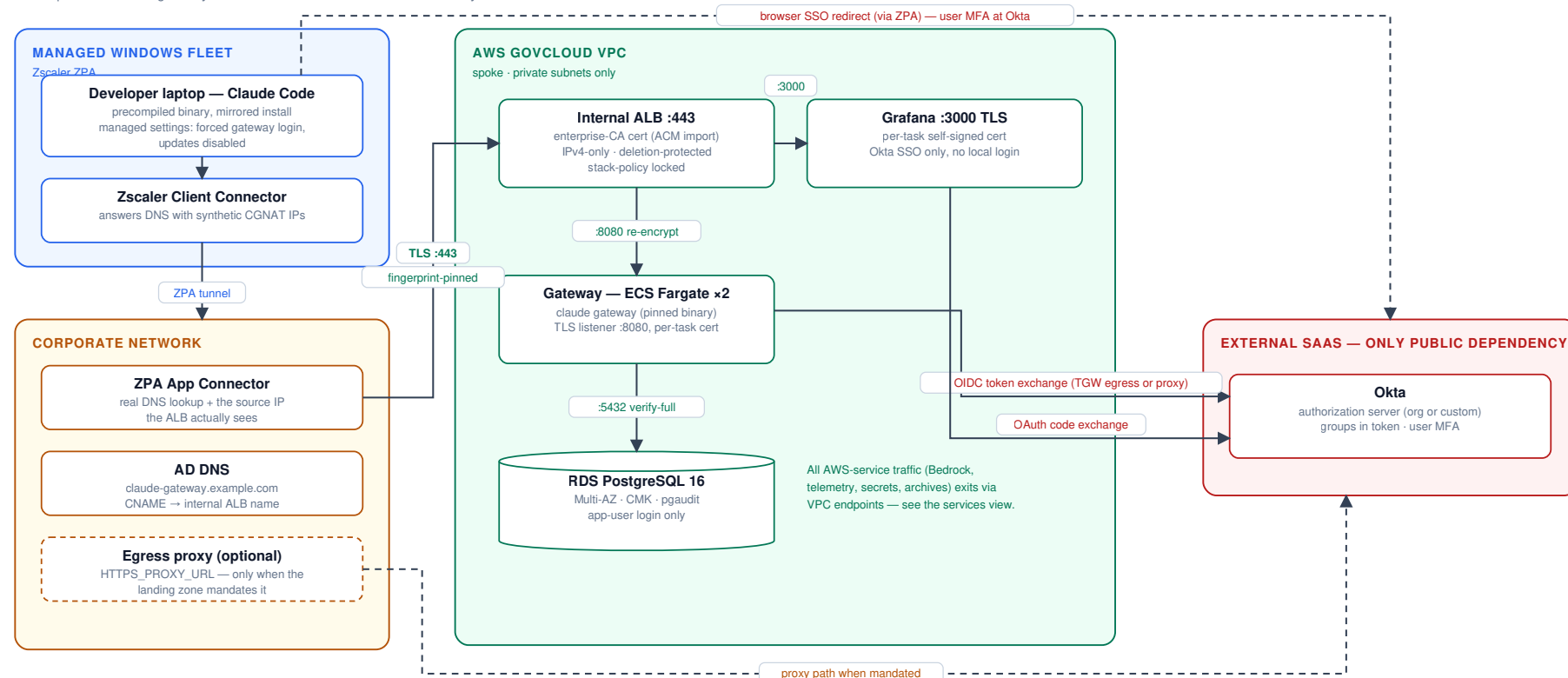
1. System architecture & trust boundaries

Four trust zones: the managed Windows fleet (Zscaler ZPA), the corporate network (App Connectors + AD DNS + egress proxy), the AWS GovCloud VPC (all workload components), and external SaaS (Okta — the single public dependency; Bedrock and all other AWS services are reached over private endpoints or the AWS backbone). Two complementary views; the combined single-page version is in the [appendix](#).

Access & network path — how requests reach the gateway and how the OIDC flows leave:

Access & network path

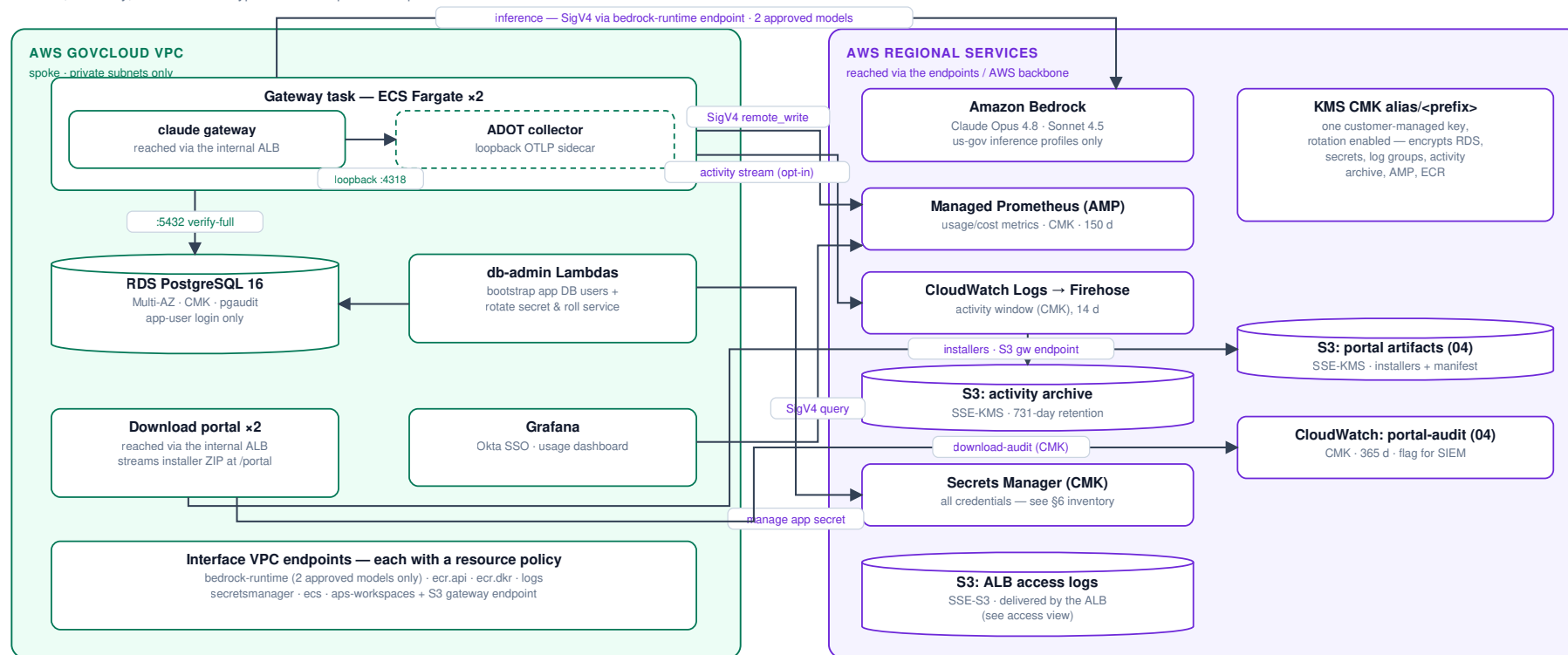
how requests reach the gateway and how the two OIDC flows leave — every arrow is TLS



Workloads & AWS data services — inference, telemetry, secrets and encryption inside AWS:

Workloads & AWS data services

inference, telemetry, secrets and encryption — developer access path is in the access view



The ADOT collector is a co-resident loopback sidecar in the gateway task (OTLP over 127.0.0.1), so telemetry never crosses the network in the clear — the former plaintext OTLP hop is gone (C2 closed). Okta egress and the developer path are on the access view.

Secrets are injected into tasks at launch by ECS (execution roles hold GetSecretValue + kms:Decrypt on exactly their own secrets — see §6).

Download portal (04): serves the installer ZIP from the CMK-encrypted artifacts bucket via the S3 gateway endpoint and writes a download-audit stream; its Okta OIDC egress is on the access view.

Key boundary statements for the reviewer:

- **No public ingress.** The ALB is internal; reaching it requires the ZPA app segment (or VPN) and an allow-listed connector source CIDR.
- **No public egress from the inference path.** Bedrock is reached via an interface endpoint whose policy allows exactly the two approved models; WebSearch is disabled on gateway sessions.
- **One public dependency:** the Okta issuer (OIDC has no VPC endpoint). Egress rides TGW central NAT or the mandated corporate proxy, port-scoped by security groups.

- **Client fleet never contacts Anthropic:** binaries are mirrored, checksum- and GPG-verified, and distributed from an internal share; update paths are disabled via the installer's user settings (and optionally the gateway's central push), and the mirror-only network path is the backstop.
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2. Network flows, ports & TLS state

Every hop, its port, protocol, and where the TLS session terminates.

Network flows, ports & TLS state — every hop in the system

one row per flow; row 6 is an in-task loopback hop (never on the network) · rows 11–15 are the download portal (stack 04)



TLS termination points: developer→ALB terminates on the enterprise cert (developers pin its fingerprint); ALB→task hops terminate on per-task ephemeral certs (ALBs do not validate target certs; keys never leave the task).

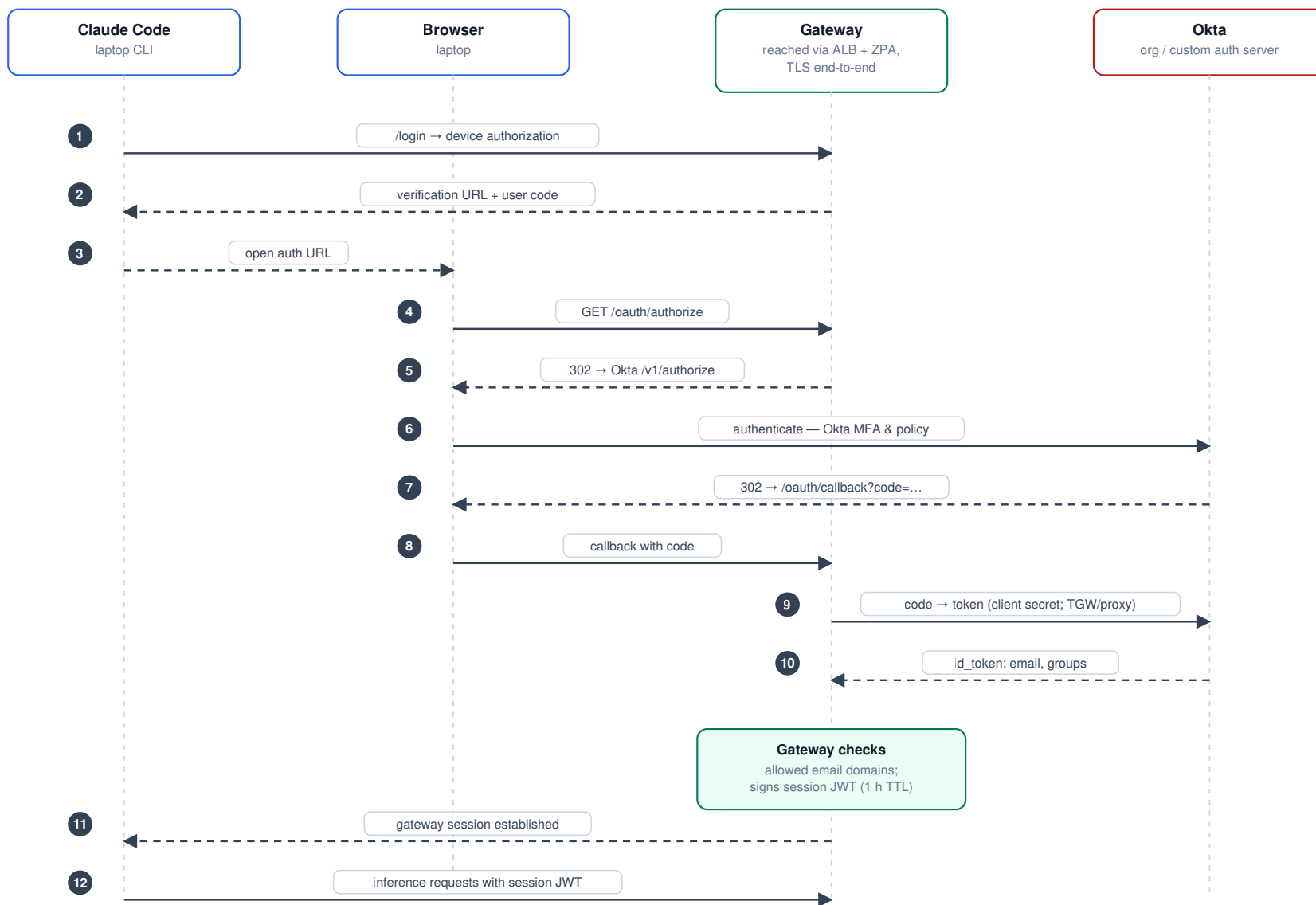
All AWS-service hops (5, 7, 8, 14, 15) also carry SigV4 request signing on top of TLS. DNS to the VPC resolver is exempt from security-group evaluation (AWS platform behavior).

#	Hop	Port	Encryption	Server identity proven by
1	Laptop → ALB	443	TLS (FIPS policy)	Enterprise-CA cert; client pins SHA-256 fingerprint on first connect
2	ALB → gateway	8080	TLS	Per-task ephemeral cert (ALB does not validate targets; key never leaves task)
3	ALB → Grafana	3000	TLS	Per-task ephemeral cert (same model)
4	Gateway/Lambda → RDS	5432	TLS <code>verify-full</code>	RDS regional CA bundle baked into images; <code>rds.force_ssl=1</code> server-side
5	Gateway → Bedrock	443	TLS + SigV4	AWS cert; endpoint policy = 2 approved models only
6	Gateway → collector (sidecar)	4317–4318	Loopback — no network hop	Same Fargate task; the collector is a localhost sidecar with its receiver bound to <code>127.0.0.1</code> , so nothing off-task can reach it (see §10)
7–8	Gateway sidecar / Grafana → AMP	443	TLS + SigV4	AWS cert; endpoint policy = this workspace only
9–10	Gateway/Grafana → Okta	443	TLS	Public CA; only internet-bound flow

3. Developer authentication (Okta OIDC)

Developer authentication — Okta OIDC via the gateway

managed settings force gateway login; no local API keys exist



Audit identity chain: Okta identity → session JWT → user.id / user.email / user.groups stamped on every request and telemetry export. Grafana repeats this flow with its own client + strict group→role mapping.

Grafana uses the **same issuer** with its own client and redirect URI (`/grafana/login/generic_oauth`), strict Okta-group→role mapping (`grafana-admins` → Admin; users in no mapped group are denied), PKCE, and the local login form disabled (break-glass `admin` requires a redeploy with `GRAFANA_DISABLE_LOGIN_FORM=false`).

Identity chain for audit: Okta identity → gateway session JWT → every inference request and every telemetry export carries `user.id/user.email/user.groups` . Per-source-IP attribution is deliberately not relied on (ZPA collapses all users to connector IPs — documented in the README).

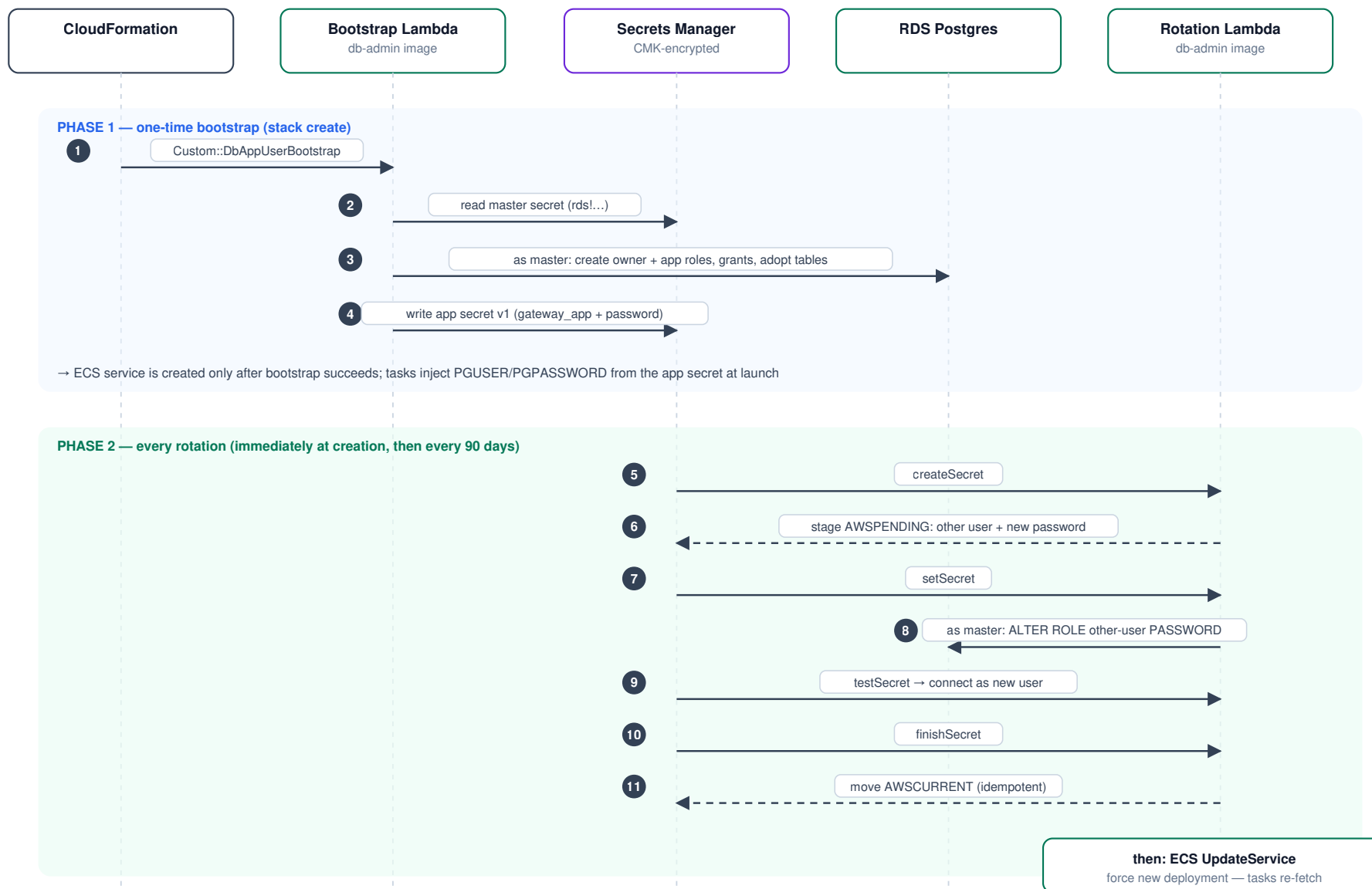
4. Database credential lifecycle

The gateway never holds the RDS master credential (AC-6). Identities:

Postgres role	Kind	Powers	Used by
<code>gw</code> (master)	LOGIN, RDS-managed	<code>rds_superuser</code> -adjacent	Break-glass humans + db-admin Lambda only
<code>gateway_owner</code>	NOLOGIN	Owns schema objects; CONNECT/TEMPORARY/CREATE on the gateway DB only	Assumed at login by both app users
<code>gateway_app</code> / <code>gateway_app_clone</code>	LOGIN (alternating pair)	Via <code>SET role</code> → <code>gateway_owner</code> ; no CREATEROLE, no rds_superuser, cannot alter pgaudit	Gateway tasks (exactly one is live)

Database credential lifecycle — least privilege + alternating-users rotation

the gateway never holds the master credential; the master secret is break-glass only



Why alternating users: the PREVIOUS credential stays valid until the NEXT rotation, so running tasks never hold a dead password — the roll is never time-critical.

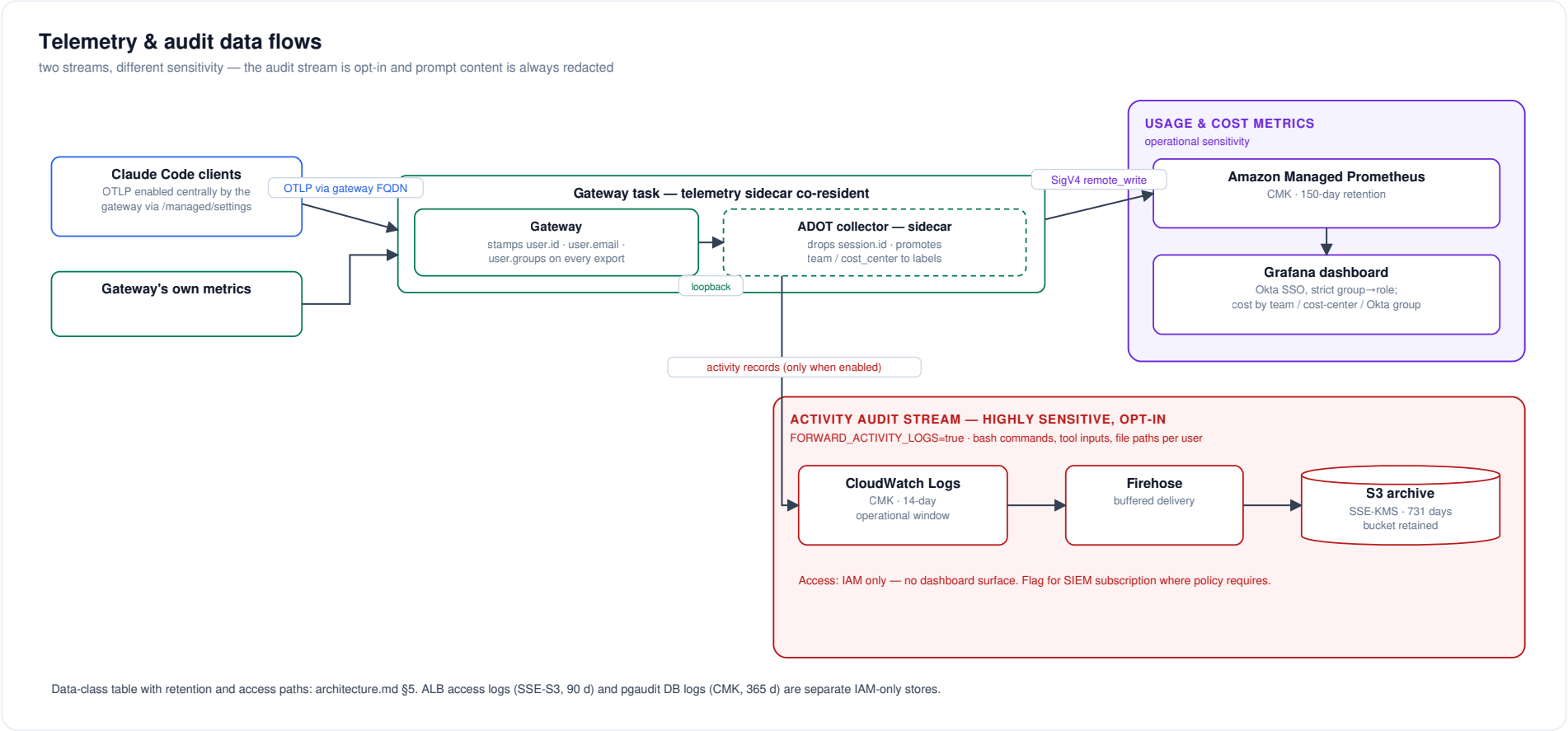
Failure handling: Secrets Manager retries any failed step; a persistently failing rotation trips the <prefix>-db-rotation-errors CloudWatch alarm.

Roles: gw (master, break-glass, RDS-rotated weekly) · gateway_owner (NOLOGIN, owns schema) · gateway_app / gateway_app_clone (LOGIN, SET role → owner; no CREATEROLE, no rds_superuser, cannot touch pgaudit).

Failure containment: a failed rotation step is retried by Secrets Manager; a persistently failing rotation trips the `<prefix>-db-rotation-errors` CloudWatch alarm. The RDS master secret keeps its RDS-managed 7-day auto-rotation, which affects no running workload.

5. Telemetry & audit data flows

Two distinct streams with different sensitivity:



Data class	Contains	Store	Retention	Access path
Usage metrics	tokens, cost, sessions, LoC, model, user identity, team/cost-center	AMP (CMK)	150 d (AMP fixed)	Grafana via Okta SSO, role-mapped
Activity stream (opt-in)	bash commands, tool inputs, file paths, per user; prompt content redacted	CloudWatch (CMK) → S3 (SSE-KMS)	14 d window / 731 d archive	IAM only; flagged for SIEM subscription
ALB access logs	source connector IPs, URLs, timings	S3 (SSE-S3)	90 d	IAM only
DB audit (pgaudit)	DDL, role changes, writes (no bind values)	RDS → CloudWatch (CMK)	365 d group	IAM only
Session/spend store	sessions, spend counters	RDS (CMK)	live	app DB user only

6. Secrets & key inventory

One customer-managed KMS key (`alias/<prefix>` , rotation enabled, created by the DB stack or bring-your-own) encrypts everything below except where noted.

Secret	Store / name	Created by	Consumed by	Rotation
RDS master (gw)	Secrets Manager rds!...	RDS-managed	Humans (break-glass) + db-admin Lambda	RDS-managed, 7 d auto
App DB user	<prefix>/db-app-user	Bootstrap Lambda	Gateway tasks (ECS injection at launch)	Stack's rotation Lambda, 90 d default, alternating users + service roll
Okta client secret (gateway)	<prefix>/oidc-client-secret	Stack (placeholder) → set-okta-secret.sh	Gateway tasks	Manual, in Okta + script (rolls service)
Okta client secret (Grafana)	<prefix>/grafana-oidc-client-secret	Stack (placeholder) → set-grafana-oidc-secret.sh	Grafana task	Manual, same pattern
Okta client secret (portal)	<prefix>/portal-oidc-client-secret	Stack 04 (placeholder) → set-portal-oidc-secret.sh	Portal tasks	Manual, same pattern
JWT session-signing secret	<prefix>/jwt-secret	Stack (generated)	Gateway tasks	Manual runbook: prepend → roll → remove
Portal session-signing secret	<prefix>/portal-session-secret	Stack 04 (generated)	Portal tasks	Manual (regenerate + roll)
Grafana admin password	<prefix>/grafana-admin-password	Stack (generated)	Break-glass only (login form disabled)	Manual
TLS: enterprise leaf + key	ACM import	Enterprise CA via import-enterprise-cert.sh	ALB listener	Manual re-import; DaysToExpiry alarm at 30 d
TLS: per-task certs	Generated in-container at startup	Gateway / Grafana endpoints	ALB target connections	Every task launch; keys never leave the task

Handling rules embedded in tooling: secret values never appear on argv (file:// + mode-600 temp files), never in CloudFormation parameters, and the two set-* -secret.sh scripts roll their consuming service.

7. Security-group matrix

Default egress is removed from **every** group (inline rules replace the allow-all); the table is the complete connectivity graph.

Deep dive with diagrams: [network-access-controls.md](#) maps every SG rule pair visually (diagram 7), inventories each group (diagram 8), and traces one API call through all five enforcement layers - SGs, endpoint SG, endpoint policies, IAM, KMS (diagram 9).

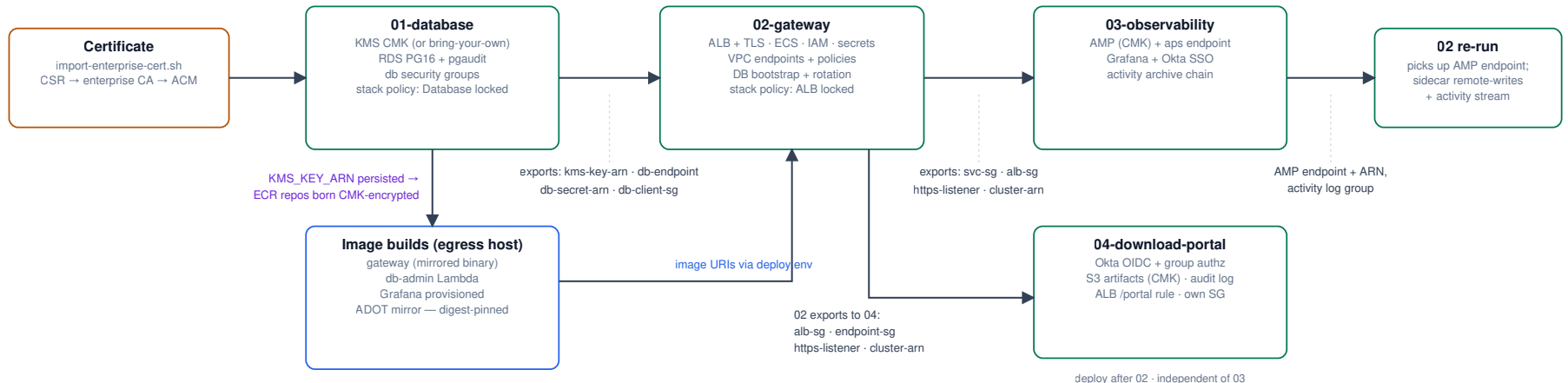
SG	Ingress	Egress
alb	443 from CLIENT_INGRESS_CIDR (ZPA connector subnets)	8080 → svc ; 3000 → grafana (added by 03); 8080 → portal (added by 04); otherwise none
svc (gateway tasks, incl. the ADOT collector sidecar)	8080 from alb	443 → 0.0.0.0/0 (Okta + AWS via endpoints/TGW; the sidecar's AMP/CloudWatch traffic rides this SG); proxy port when configured; 443 → amp-endpoint (added by 03, for the sidecar's AMP remote-write); 5432 via attached db-client
db-client (attached to tasks + db-admin Lambdas)	—	5432 → db only
db	5432 from db-client	none
db-admin (Lambdas)	—	443 → 0.0.0.0/0 (Secrets Manager, ECS APIs)
grafana	3000 from alb	443 → 0.0.0.0/0 (AMP, Okta); proxy port when configured
portal (download portal, 04)	8080 from alb	443 → 0.0.0.0/0 (Okta, S3, CloudWatch); proxy port when configured
endpoints / amp-endpoint	443 from svc / db-admin (resp. svc (gateway sidecar) + grafana + portal)	none

DNS to the VPC resolver is exempt from SG evaluation (AWS platform behavior) and is why no port-53 rules appear.

8. Stack dependencies & deploy order

Stack dependencies & deploy order

arrows carry CloudFormation exports — while imported, an export's value is locked (day-one decisions called out)



Locks a reviewer should know: the RDS storage CMK is fixed at creation (plus 01 ↔ 02 export locks) — a day-one decision; 03 and 04 must be deleted before 02 replacement-updates; the ALB and Database carry stack policies denying Update:Replace / Update>Delete.

Full teardown/update ordering: README — Teardown & update order.

Export locks (documented in README "Teardown & update order"): while a downstream stack imports an export, the upstream stack cannot change its value — most notably the RDS storage CMK is a **day-one decision**, and 03 must be deleted before 02 replacement-updates. The ALB and Database additionally carry CloudFormation **stack policies** denying `Update:Replace` / `Update>Delete`, so an accidental template change that would recreate them (new ALB DNS name → client DNS resubmission; new empty database) fails fast.

9. Data-at-rest encryption map

Store	Encryption	Key
RDS storage + snapshots	SSE	CMK
RDS master secret / all Secrets Manager secrets	SSE	CMK
CloudWatch log groups (ECS gateway [+ its collector sidecar, stream prefix <code>otel</code>], grafana/portal, activity, portal-audit, RDS postgresql/pgaudit, db-admin Lambdas)	SSE	CMK — every group is template-declared (never service-auto-created) precisely so the CMK and retention apply; all carry <code>DeletionPolicy: Retain</code> (operator decision 2026-07-18: no log group is destroyed by a teardown)
Activity archive bucket	SSE-KMS + bucket key	CMK
Portal artifacts bucket (04)	SSE-KMS + bucket key	CMK
AMP workspace	SSE	CMK (<code>ENCRYPT_AMP_WITH_CMK</code> , creation-time)
ECR repositories	SSE-KMS at creation	CMK (when created after 01)
ALB access-logs bucket	SSE-S3 (AES-256)	AWS-managed — ELB log delivery does not support KMS

10. Known accepted risks / review notes

Items a reviewer should see up front, with rationale (full history in `security-review-2026-07.md`):

1. **Gateway→collector OTLP hop — formerly plaintext-but-SG-scoped, now eliminated** (§2 hop 6). This was an accepted partial risk; the live run withdrew it. The gateway refuses a non-HTTPS telemetry forward URL unless the host is localhost, so the plaintext network hop could never boot, and an internal-CA TLS listener would add a cert + CA the gateway must trust (an SC-17/SC-12 burden the org declined). The collector was therefore moved into the gateway task as a **localhost sidecar** (2026-07-22): the gateway forwards over loopback (`http://localhost:4318`), so there is no network telemetry leg and SC-8 is met by absence of transmission. The gateway task role is now the telemetry writer to AMP + the activity log group (scoped to this workspace and this log group). Pending live verification that metrics reach AMP through the sidecar (`security-review-2026-07.md` C2 + fix log).
2. **ALB access logs cannot use the CMK** — AWS platform limitation; SSE-S3 with public access blocked and lifecycle expiry.
3. **S3 Object Lock deferred** by decision (2026-07-15); revisit if AU-9 WORM retention is mandated.
4. **Egress 443 to 0.0.0.0/0 from tasks** — required for Okta (public SaaS, no VPC endpoint, IP ranges not pinnable); mitigated by port scoping, endpoint policies for all AWS traffic, and the central inspection layer in the landing zone.
5. **Foundation-model IAM region wildcard** (`bedrock:*::foundation-model/ <exact-model>`) — required because GovCloud geo inference profiles fan out across us-gov regions; model IDs are exact.

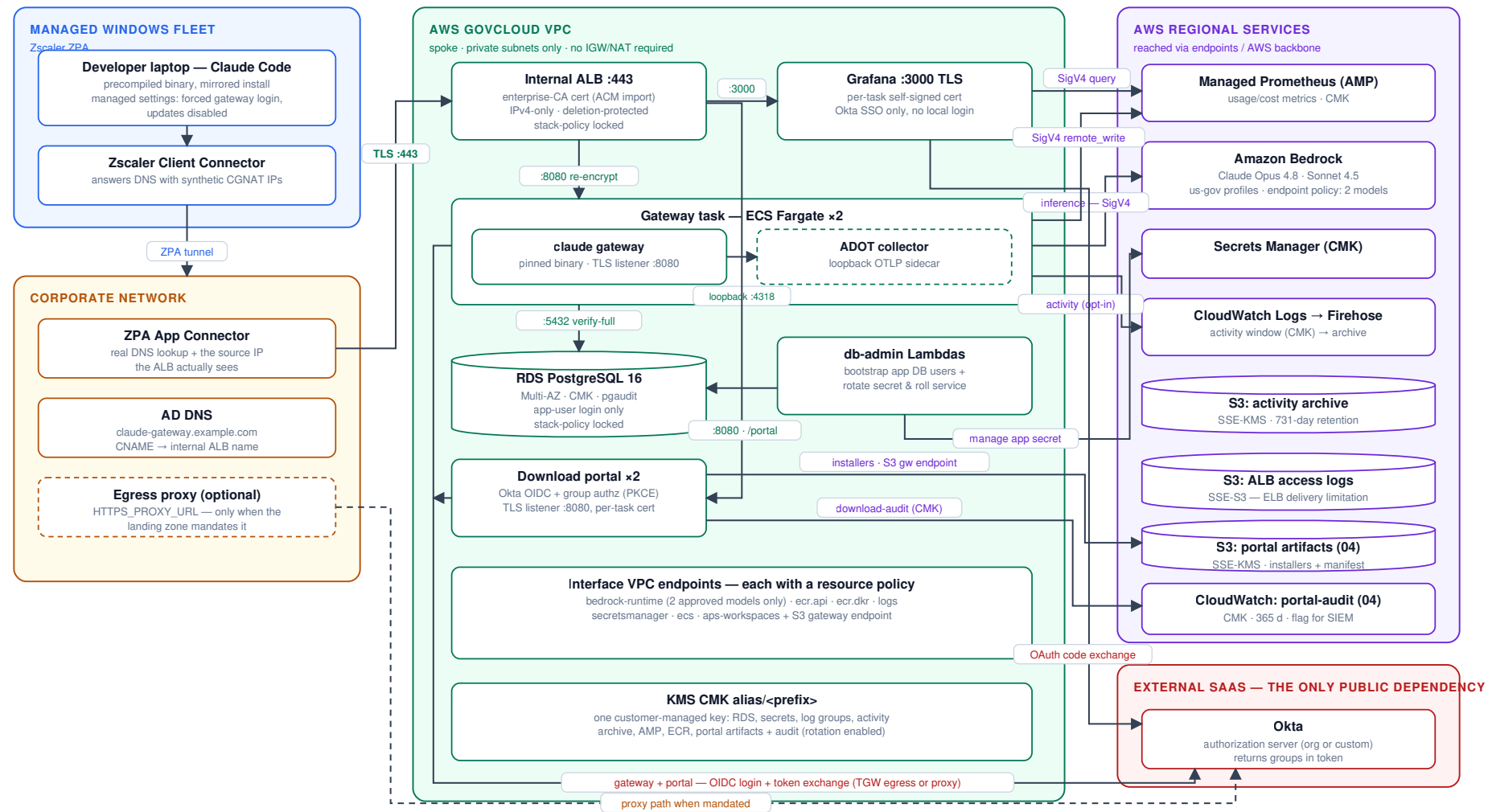
6. **First app-secret rotation is asynchronous** — the stack goes green regardless; verify via the db-rotation log group / errors alarm after first deploy.

Appendix — combined system view

Everything from §1 on a single page — denser than the split views above, useful when a reviewer wants the whole system at a glance.

System architecture & trust boundaries

Claude apps gateway in AWS GovCloud us-gov-west-1 — every arrow is TLS unless flagged



Boundary facts: no public ingress (internal ALB behind ZPA); no public egress from the inference path (Bedrock endpoint, 2-model policy); clients never contact Anthropic (mirrored, verified binaries; updates disabled).